

01 Vectors



1.6 Vector Inner Product

From summing paired products to length times cosine — two views of the same idea

Linear Algebra Made Easy and Visual | Open-source code & drafts: <https://github.com/Visualize-ML>

What You Will Learn

1

Master the algebraic definition of the inner product: $\mathbf{a} \cdot \mathbf{b} = \sum a_i b_i$, pairwise products summed into a scalar

2

Master its algebraic properties: commutativity, distributivity, and scalar associativity

3

Understand its geometric meaning: $\mathbf{a} \cdot \mathbf{b} = \|\mathbf{a}\| \|\mathbf{b}\| \cos \theta$, length times the cosine of the angle

4

Use the inner product to test orthogonality: zero inner product $\Leftrightarrow$ perpendicular vectors; distinguish orthogonal from orthonormal vectors

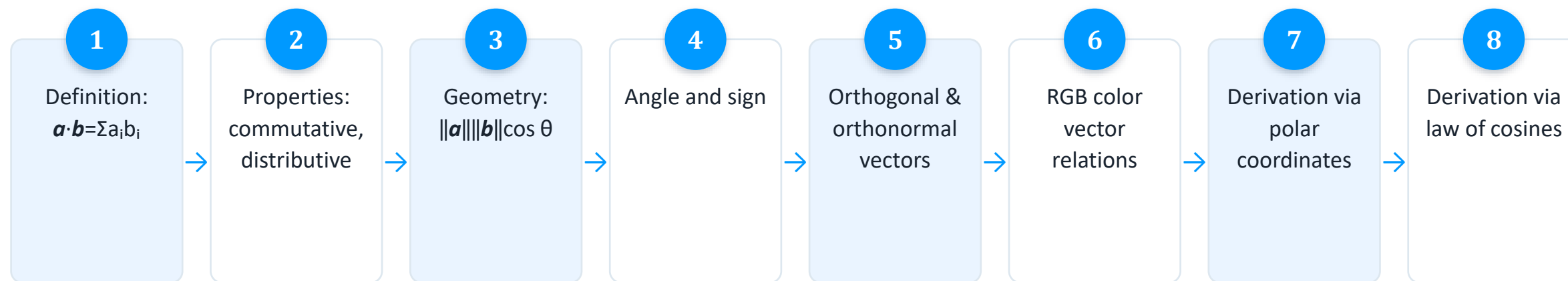
5

Understand how angle relates to similarity, and apply it to color vectors in RGB space

6

Derive the inner product formula from both polar coordinates and the law of cosines, and compute it with NumPy

How the Ideas Connect



Inner Product: Multiply, Then Sum

$$\mathbf{a} \cdot \mathbf{b} = \sum_{i=1 \rightarrow n} a_i b_i$$

- For column vectors $\mathbf{a}$, $\mathbf{b}$ of the same dimension, their inner product sums the products of matching components into a single scalar
- Σ is the summation symbol; $i=1$ marks the starting index, accumulating up to $i=n$; $a_i b_i$ is the product of the i -th components
- Also called the scalar product or dot product; this book writes it with a solid dot $\cdot$, which should never be dropped

A Worked Example

$$\mathbf{a}=[1,2,3]^T, \mathbf{b}=[4,5,6]^T \Rightarrow \mathbf{a} \cdot \mathbf{b} = 1 \times 4 + 2 \times 5 + 3 \times 6 = 32$$

- Multiply matching components: $1 \times 4 = 4$, $2 \times 5 = 10$, $3 \times 6 = 18$
- Sum the three products: $4 + 10 + 18 = 32$ — a scalar, not a vector
- Also written as $\langle \mathbf{a}, \mathbf{b} \rangle$ or $\mathbf{a}^T \mathbf{b}$; all these notations mean the same thing in this book

Computing the Inner Product with NumPy

np.dot(a_vec, b_vec) returns the inner product in one call

- Define vectors `a_vec=[1,2,3]` and `b_vec=[4,5,6]` with `np.array()`
- `a_dot_b = np.dot(a_vec, b_vec)` computes the inner product directly, giving 32
- Check: `a_vec * b_vec` multiplies elementwise, and `sum(a_vec * b_vec)` sums the results — both agree

```
## 初始化
import numpy as np

## 定义两个向量
a_vec = np.array([1, 2, 3])
b_vec = np.array([4, 5, 6])

## 计算内积
a_dot_b = np.dot(a_vec, b_vec)

## 验算
a_vec * b_vec
sum(a_vec * b_vec)
```



Three Everyday Properties of the Inner Product

$$\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a}; \mathbf{a} \cdot (\mathbf{b} + \mathbf{c}) = \mathbf{a} \cdot \mathbf{b} + \mathbf{a} \cdot \mathbf{c}; k(\mathbf{a} \cdot \mathbf{b}) = (k\mathbf{a}) \cdot \mathbf{b} = \mathbf{a} \cdot (k\mathbf{b})$$

- Commutative: order does not matter — $\mathbf{a} \cdot \mathbf{b}$ equals $\mathbf{b} \cdot \mathbf{a}$
- Distributive: the inner product distributes over vector addition, just like ordinary multiplication
- Scalar associative: a scalar k can move freely across the inner product, where $\mathbf{a}, \mathbf{b}, \mathbf{c}$ share the same dimension

A Vector With Itself, and With the All-Ones Vector

$$\mathbf{a} \cdot \mathbf{a} = \|\mathbf{a}\|^2 = \sum a_i^2; \mathbf{a} \cdot \mathbf{1} = \sum a_i$$

- The inner product of $\mathbf{a}$ with itself equals the sum of its squared components — the squared length $\|\mathbf{a}\|^2$
- Clearly $\mathbf{a} \cdot \mathbf{a} \geq 0$, with equality if and only if $\mathbf{a}$ is the zero vector
- If one of the two vectors is the all-ones vector $\mathbf{1}=[1,1,\dots,1]^T$, the result is simply the sum of $\mathbf{a}$'s components

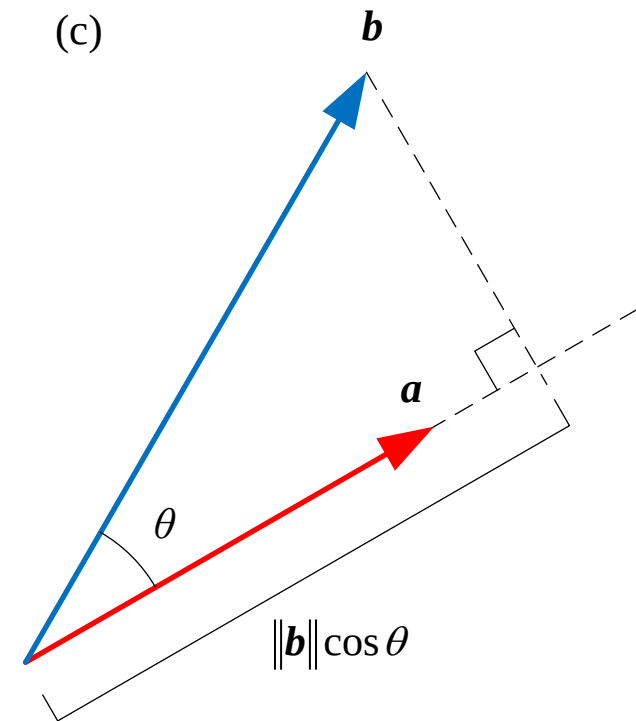
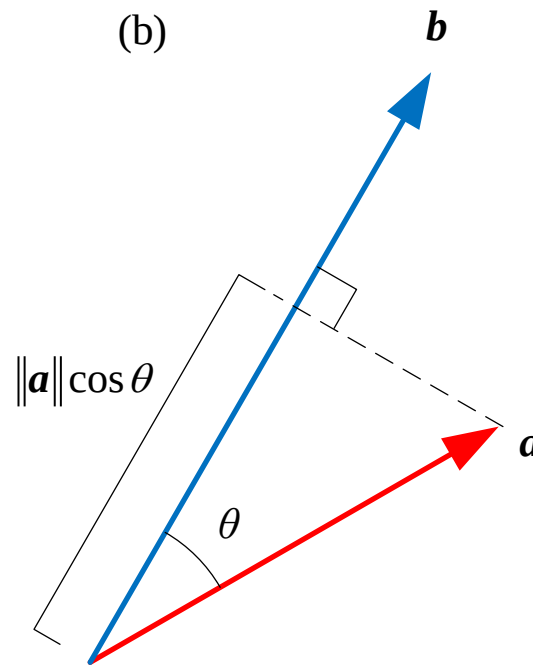
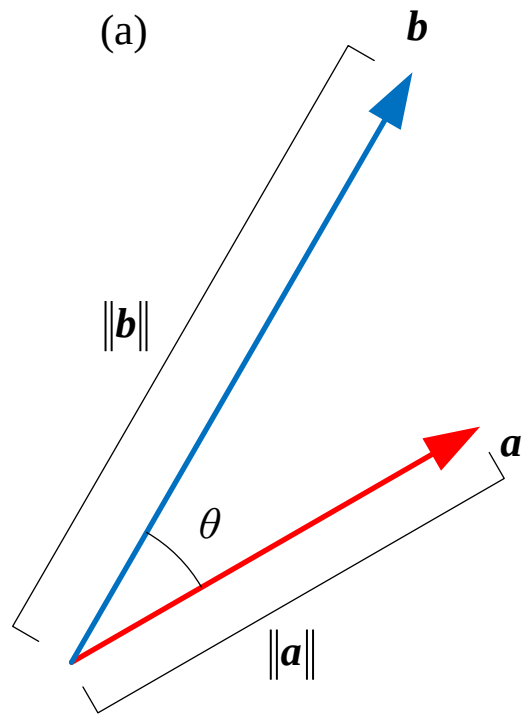
Geometric Meaning: Length Times the Cosine of the Angle

$$\mathbf{a} \cdot \mathbf{b} = \|\mathbf{a}\| \|\mathbf{b}\| \cos \theta$$

- Geometrically, the inner product equals the product of the two lengths, scaled by the cosine of the angle θ between them
- θ is the angle between $\mathbf{a}$ and $\mathbf{b}$, usually taken in the range $[0^\circ, 180^\circ]$
- This formula ties the algebraic quantity directly to two geometric ones: length and angle

Figure 1 · The Inner Product, Seen Geometrically

- (a) θ is the angle between $\mathbf{a}$ and $\mathbf{b}$; $\|\mathbf{a}\|$ and $\|\mathbf{b}\|$ are their lengths
- (b)(c) $\|\mathbf{a}\|\cos\theta$ and $\|\mathbf{b}\|\cos\theta$ are the scalar projections of $\mathbf{a}$ onto $\mathbf{b}$, and of $\mathbf{b}$ onto $\mathbf{a}$



How the Angle θ Sets the Sign and Size of the Inner Product

$\theta = 0^\circ$ $\mathbf{a}$ and $\mathbf{b}$ point the same way; inner product is maximal,
 $\cos(0^\circ)=1$

$0^\circ < \theta < 90^\circ$ Acute angle — the inner product is positive

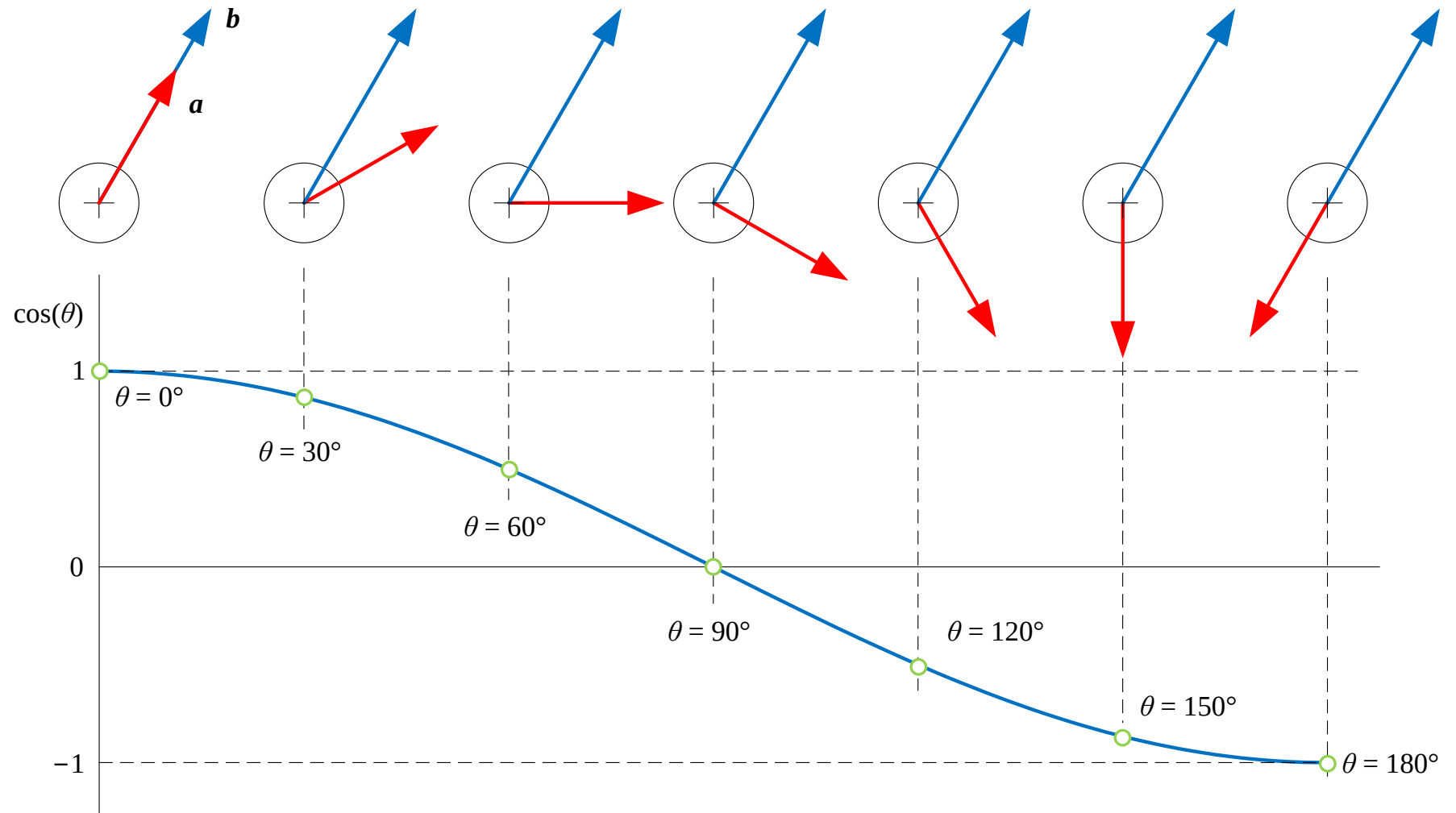
$\theta = 90^\circ$ $\mathbf{a}$ and $\mathbf{b}$ are orthogonal; inner product is 0, $\cos(90^\circ)=0$

$90^\circ < \theta < 180^\circ$ Obtuse angle — the inner product is negative

$\theta = 180^\circ$ $\mathbf{a}$ and $\mathbf{b}$ point opposite ways; inner product is minimal,
 $\cos(180^\circ)=-1$

Figure 2 · The Inner Product as θ Varies

- $\mathbf{a}$ and $\mathbf{b}$ keep fixed lengths while only θ changes; the small diagrams above show each orientation
- The curve below traces $\cos \theta$: maximal at $\theta=0^\circ$, zero at $\theta=90^\circ$, minimal at $\theta=180^\circ$



Recovering the Cosine of the Angle From the Inner Product

$$\cos \theta = \mathbf{a} \cdot \mathbf{b} / (\|\mathbf{a}\| \|\mathbf{b}\|)$$

- For nonzero $\mathbf{a}$ and $\mathbf{b}$, the geometric definition can be rearranged to solve directly for $\cos \theta$
- This is equivalent to taking the inner product of the two direction vectors (unit vectors), which carry direction but not length
- The sign of $\cos \theta$ tells you at a glance whether the angle is acute, right, or obtuse — see the previous slide

Computing the Angle Between Two Vectors

`np.arccos()` and `np.degrees()` recover the angle

- Define `vector1=[1,1,0]` and `vector2=[0,1,1]` with `np.array()`
- `np.dot(vector1, vector2)` computes the inner product; `np.linalg.norm()` computes each vector's length
- `cos_theta = dot_product/(norm1*norm2)` recovers the cosine of the angle from the inner product and the lengths
- `np.arccos()` gives the angle in radians, and `np.degrees()` converts it to degrees

```
## 初始化
import numpy as np

## 定义两个向量
a vector1 = np.array([1, 1, 0])
  vector2 = np.array([0, 1, 1])

## 计算内积
b dot_product = np.dot(vector1, vector2)

## 计算向量长度
c norm1 = np.linalg.norm(vector1)
  norm2 = np.linalg.norm(vector2)

## 计算夹角的余弦值
d cos_theta = dot_product / (norm1 * norm2)

## 计算夹角（弧度）
e theta_rad = np.arccos(cos_theta)

## 计算夹角（角度）
f theta_deg = np.degrees(theta_rad)
```



Orthogonal Vectors and Orthonormal Vectors

Orthogonal Vectors A set of vectors whose pairwise inner products are 0 — a 90° angle between each pair

Orthonormal Vectors A set of vectors that are pairwise orthogonal and each has length 1

RGB Basis Vectors $\mathbf{e}_1=[1,0,0]^T$, $\mathbf{e}_2=[0,1,0]^T$, $\mathbf{e}_3=[0,0,1]^T$ stand for the red, green, and blue components

Pairwise Orthogonal $\mathbf{e}_1 \cdot \mathbf{e}_2 = \mathbf{e}_1 \cdot \mathbf{e}_3 = \mathbf{e}_2 \cdot \mathbf{e}_3 = 0$, and all three have length 1 — an orthonormal set

Figure 3 · Orthonormal Vectors in RGB Space

- The red, green, and blue basis vectors $\mathbf{e}_1=[1,0,0]^T$, $\mathbf{e}_2=[0,1,0]^T$, $\mathbf{e}_3=[0,0,1]^T$ meet at 90° pairwise, with zero inner product
- $\mathbf{e}_1$, $\mathbf{e}_2$, $\mathbf{e}_3$ all have length 1, making them a mutually orthogonal set of unit vectors — orthonormal vectors

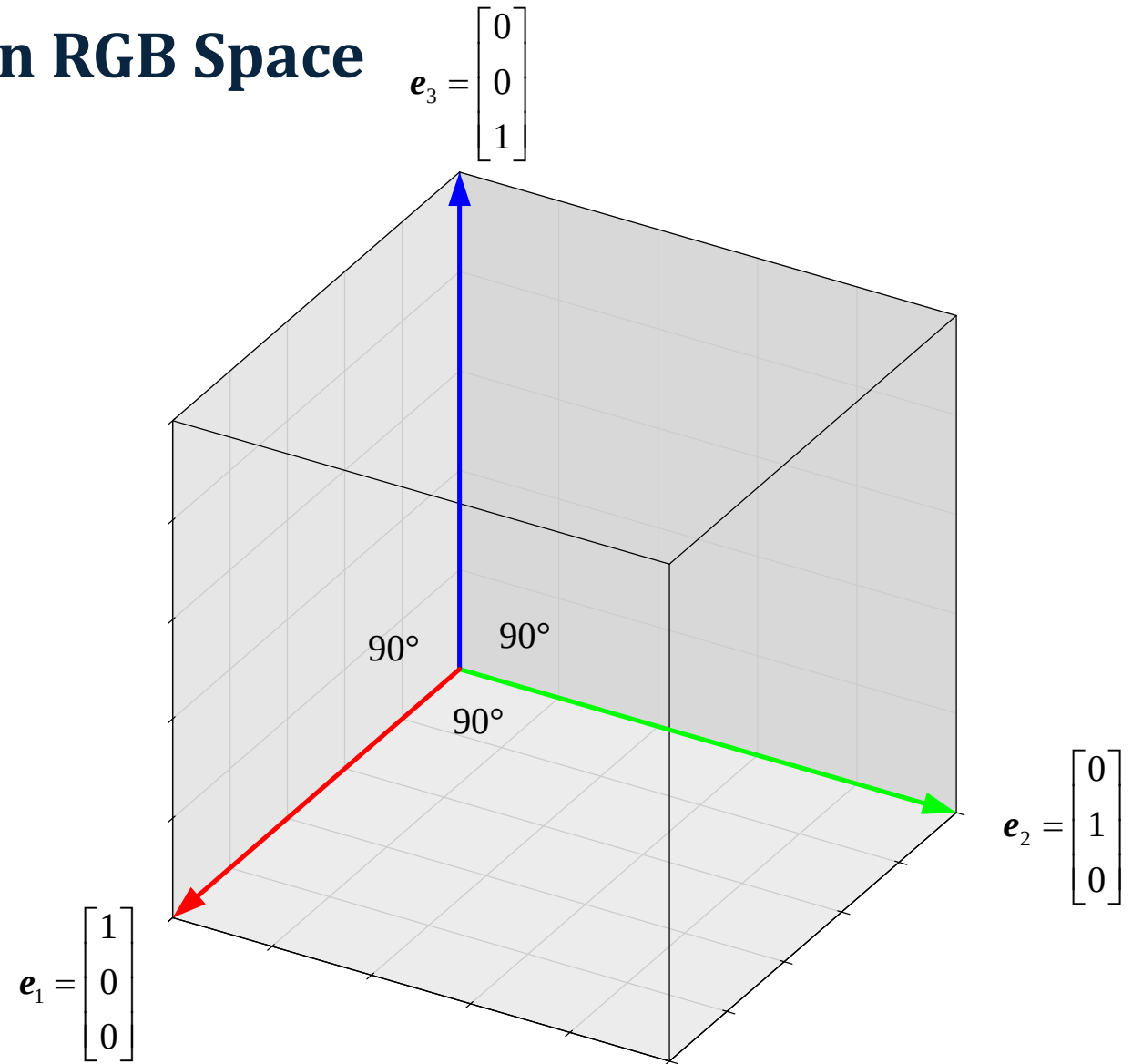


Figure 4 · Red Is Orthogonal to the Entire "Blue-Green" Plane

- The dense fan of blue-green gradient vectors are all linear combinations of blue $\mathbf{e}_3$ and green $\mathbf{e}_2$, spanning the "blue-green" plane
- The red vector $\mathbf{e}_1$ has zero inner product with every vector in that plane — it is orthogonal to the whole plane

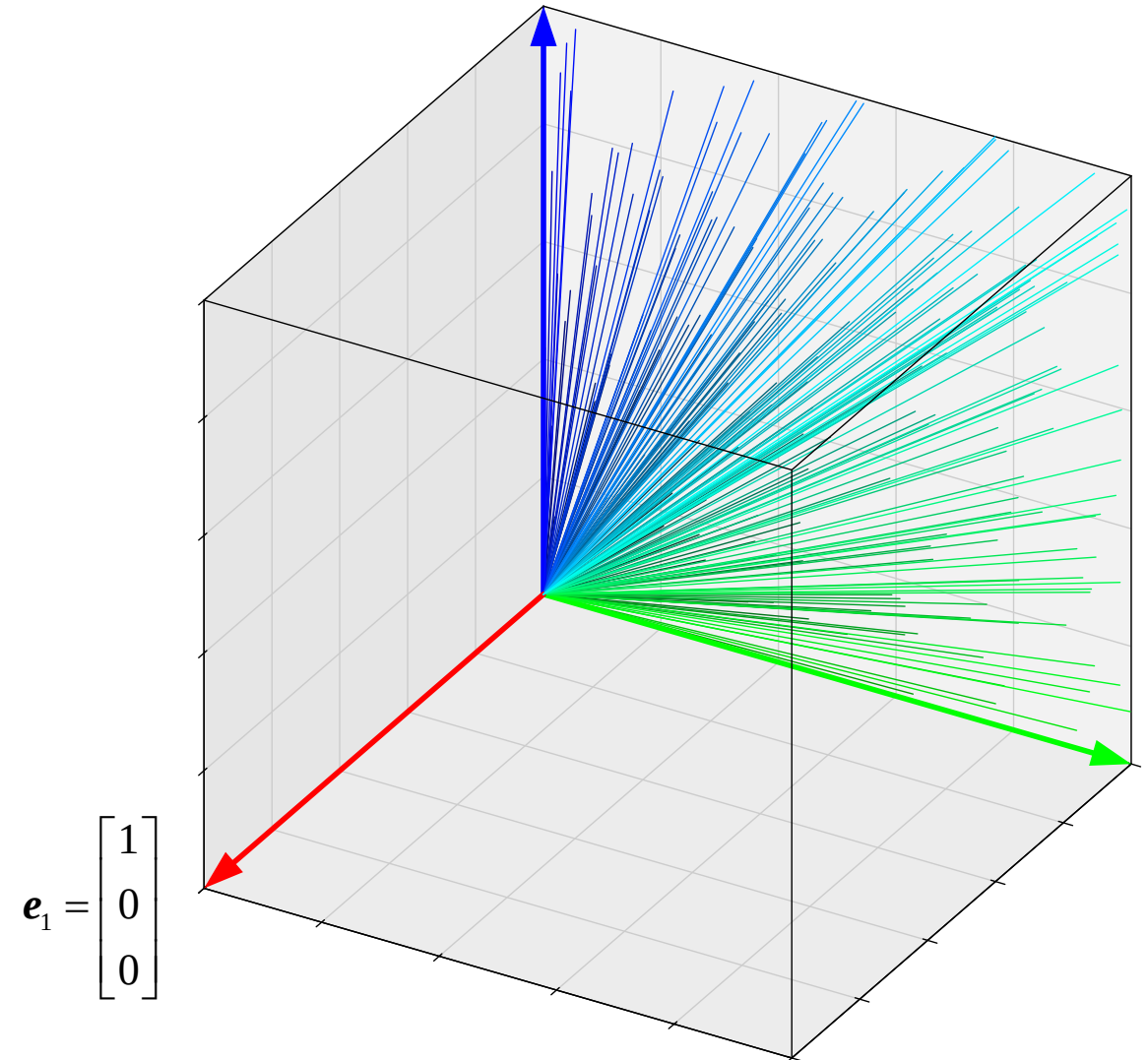
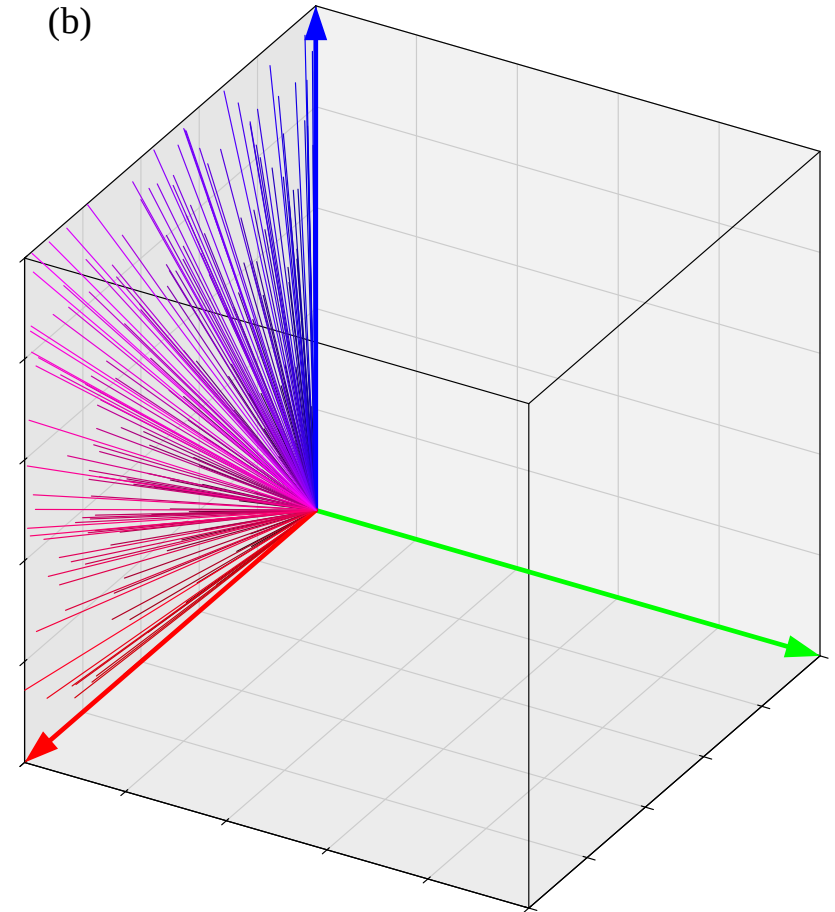
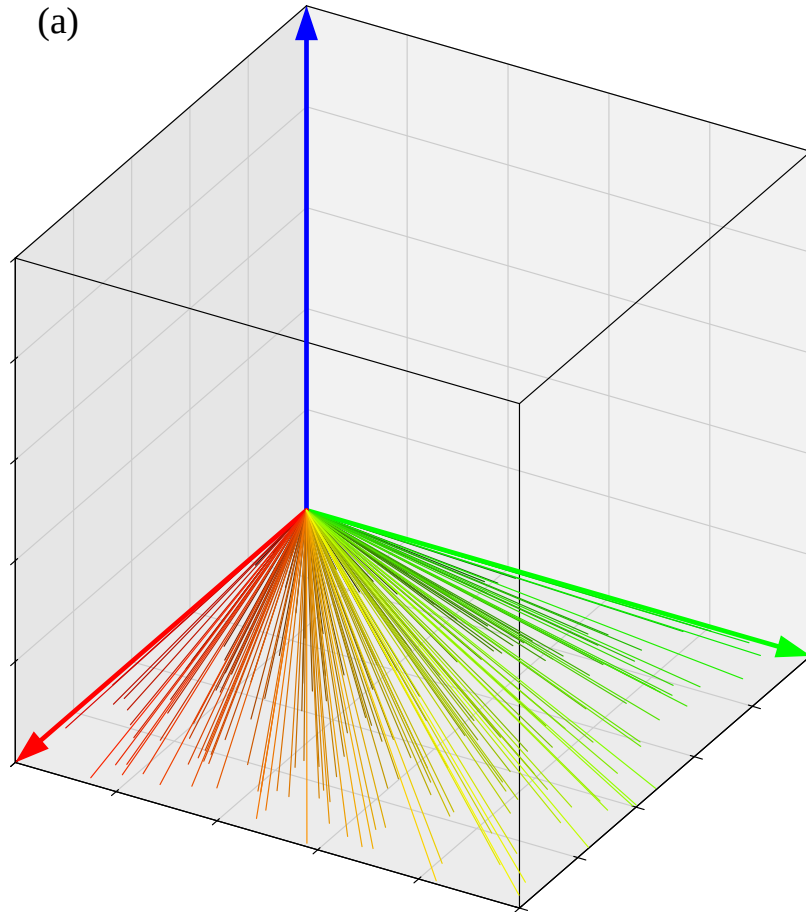


Figure 5 · Blue and Green Are Each Orthogonal to Their Plane

- (a) Blue $\mathbf{e}_3$ is orthogonal to the "red-green" plane spanned by red $\mathbf{e}_1$ and green $\mathbf{e}_2$
- (b) Green $\mathbf{e}_2$ is orthogonal to the "red-blue" plane spanned by red $\mathbf{e}_1$ and blue $\mathbf{e}_3$



Red and Yellow Meet at a 45° Angle

$$\mathbf{e}_1 \cdot (\mathbf{e}_1 + \mathbf{e}_2) = 1, \cos \theta = 1/\sqrt{2}, \theta = 45^\circ$$

- Yellow is red plus green: $\mathbf{e}_1 + \mathbf{e}_2 = [1, 1, 0]^T$ — clearly red and yellow are no longer orthogonal
- Red has length 1, yellow has length $\sqrt{2}$; their inner product is 1, so $\cos \theta = 1/\sqrt{2}$
- That gives a 45° angle between red and yellow; by the same reasoning, green and yellow also meet at 45°

Figure 6 · Color Vectors at 45° in RGB Space

- Yellow e_1+e_2 , magenta e_1+e_3 , and cyan e_2+e_3 are each combinations of two base colors
- Red and yellow meet at 45°; the white vector $e_1+e_2+e_3$ runs along the cube's diagonal

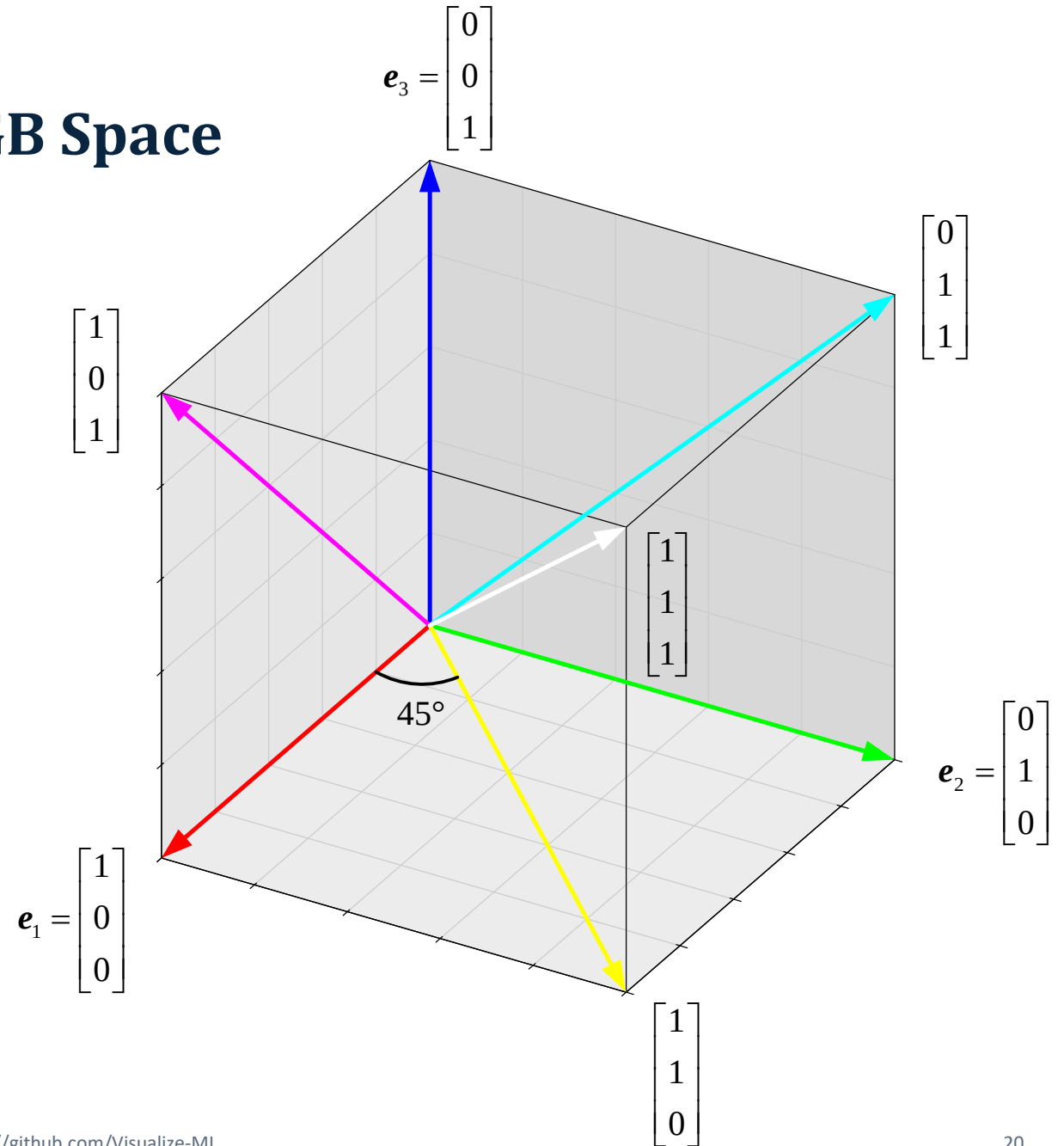
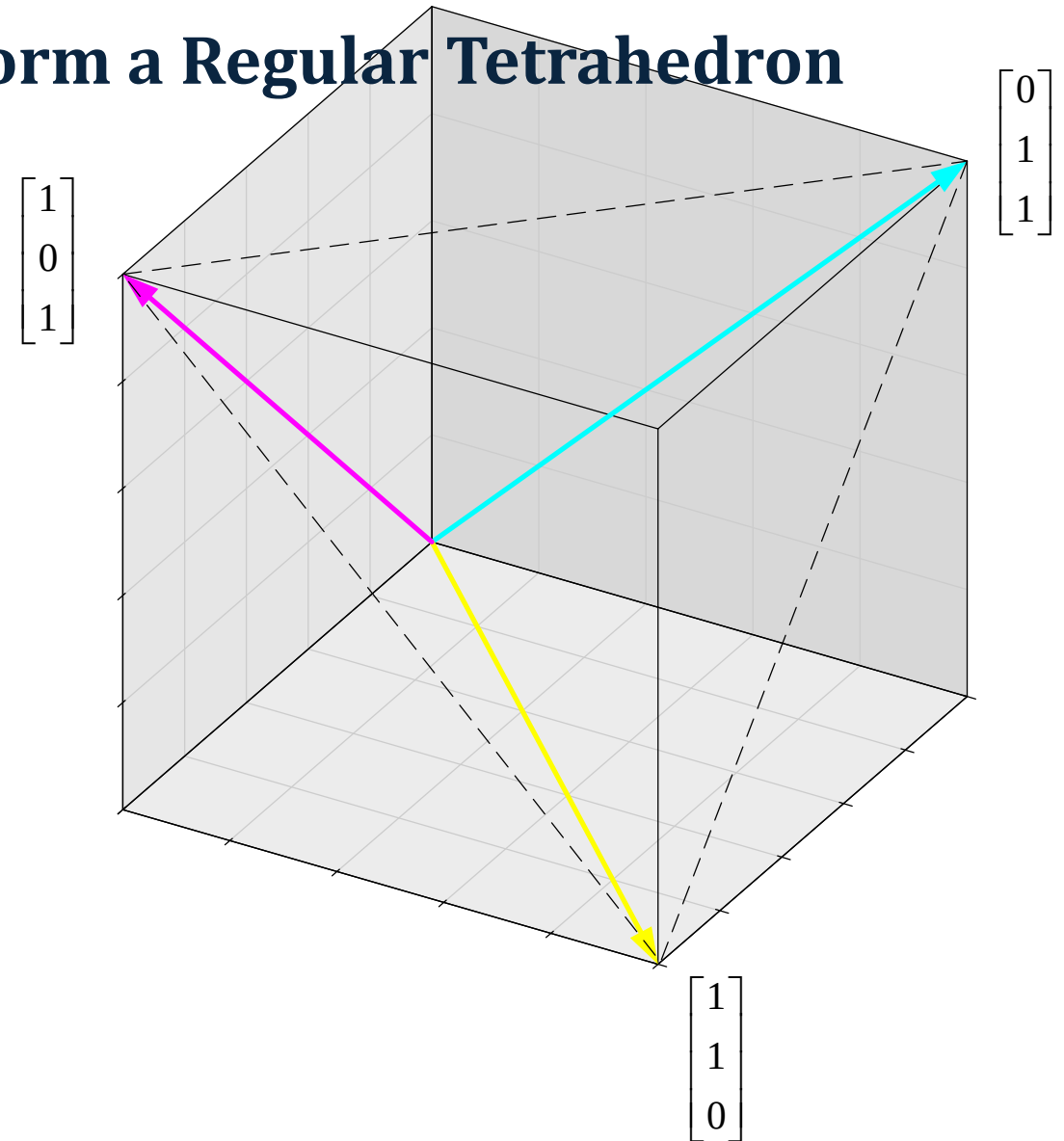


Figure 7 · Yellow, Magenta, and Cyan Form a Regular Tetrahedron

- The tips of yellow $\mathbf{e}_1 + \mathbf{e}_2$, magenta $\mathbf{e}_1 + \mathbf{e}_3$, and cyan $\mathbf{e}_2 + \mathbf{e}_3$ are equally spaced, forming an equilateral triangle
- Every pair among them meets at the same 60° angle — a symmetric relationship between these three "mixed" colors



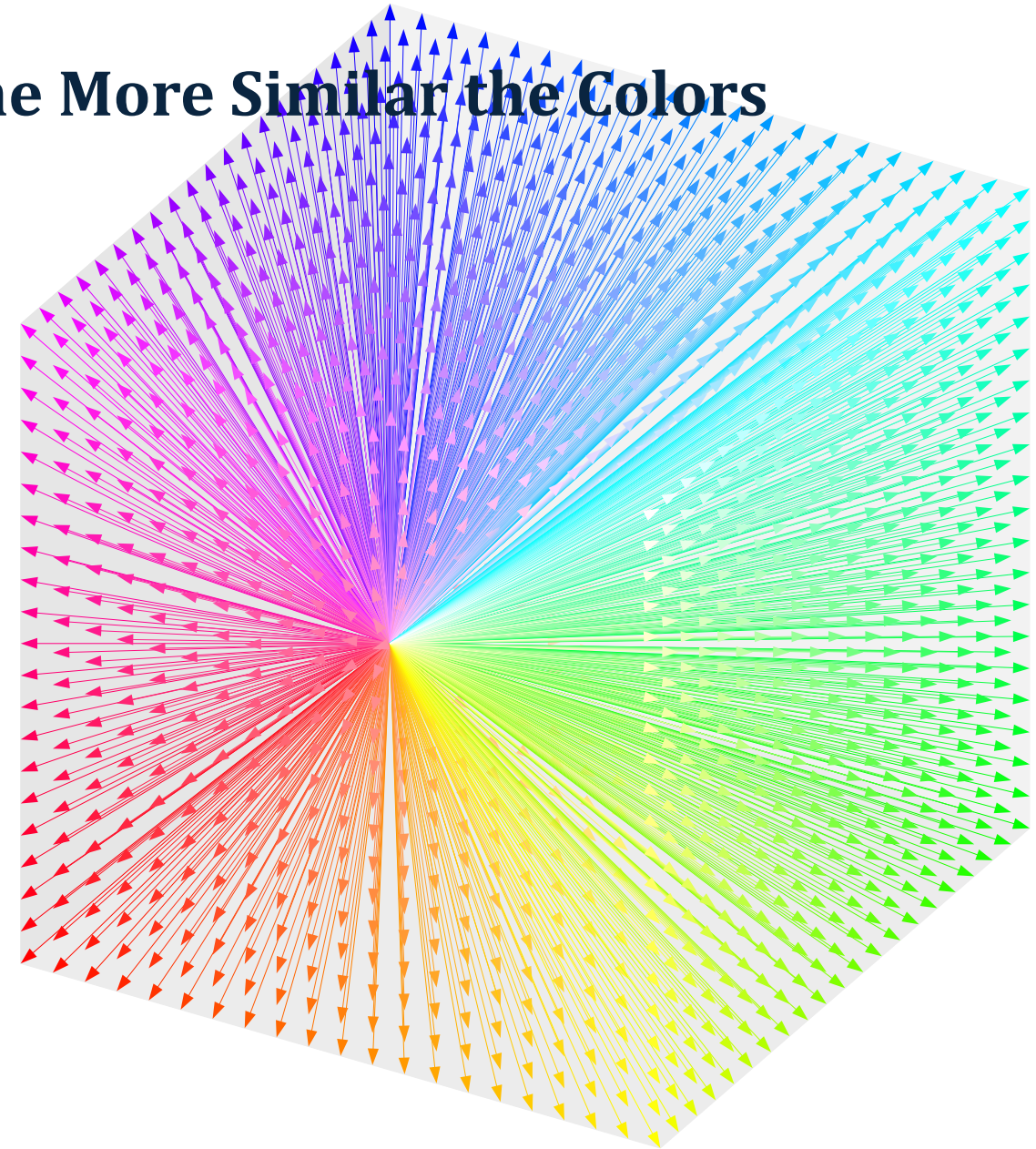
Angles Among Yellow, Cyan, and White

$$(e_1 + e_2) \cdot (e_2 + e_3) = 1, \cos \theta = 1/2, \theta = 60^\circ$$

- Yellow and cyan have inner product 1 and length $\sqrt{2}$ each, giving an angle of $\theta=60^\circ$
- White $e_1 + e_2 + e_3$ and red have inner product 1, with lengths $\sqrt{3}$ and 1 — an angle of about 54.8°
- White and yellow have inner product 2, with lengths $\sqrt{3}$ and $\sqrt{2}$ — an angle of about 35.3° , meaning more overlap

Figure 8 · The Closer the Angle to 0° , the More Similar the Colors

- Each colorful vector on the sphere stands for a color, and the angle between two vectors reflects how similar they are
- An angle near 0° means the colors are similar; an angle near 90° means they differ more, with less shared component



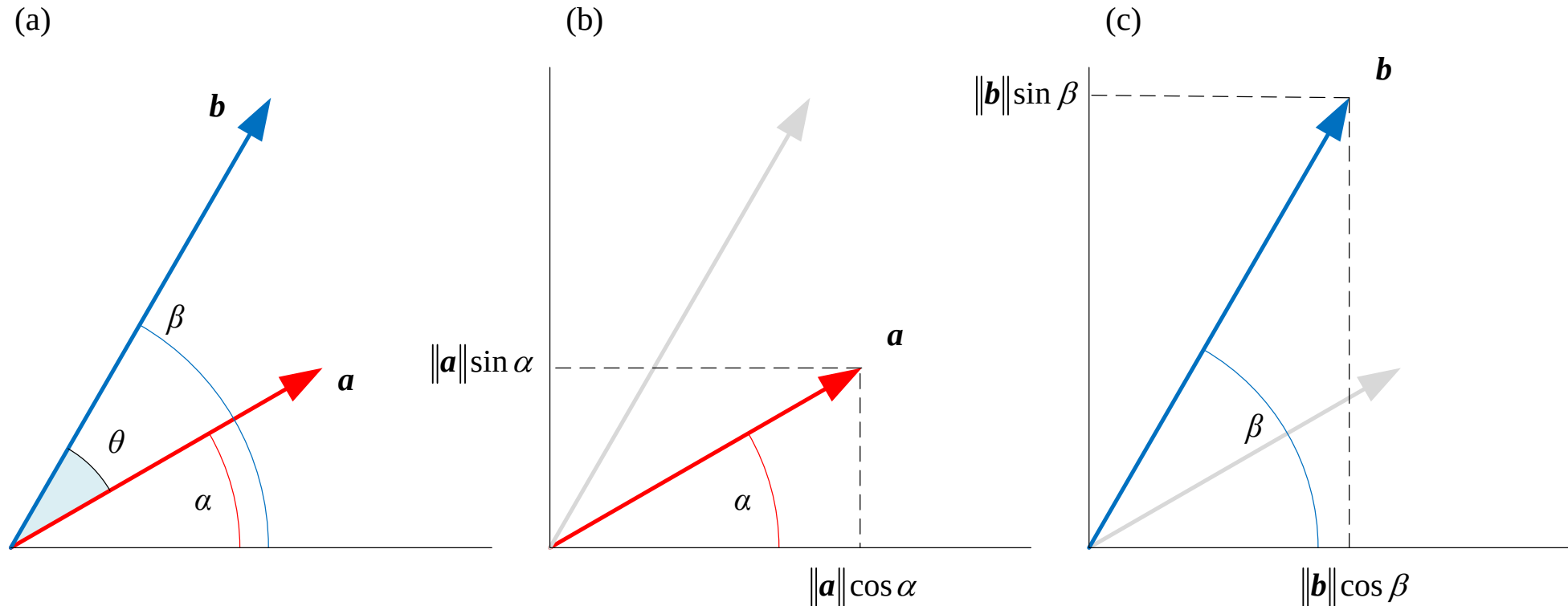
Writing $\mathbf{a}$ and $\mathbf{b}$ in Polar Coordinates

$\mathbf{a}$ has polar coordinates $(\|\mathbf{a}\|, \alpha)$; $\mathbf{b}$ has $(\|\mathbf{b}\|, \beta)$; $\theta = \beta - \alpha$

- α is the angle between $\mathbf{a}$ and the polar axis (the positive x_1 direction); β is the same for $\mathbf{b}$
- The angle between $\mathbf{a}$ and $\mathbf{b}$ is simply the difference of the two polar angles, $\beta - \alpha$
- A trigonometric identity lets us expand $\cos \theta = \cos(\beta - \alpha)$ into Cartesian-coordinate terms

Figure 9 · Vectors $\mathbf{a}$ and $\mathbf{b}$ in Polar and Cartesian Coordinates

- (a) $\mathbf{a}$ and $\mathbf{b}$ have polar coordinates $(\|\mathbf{a}\|, \alpha)$ and $(\|\mathbf{b}\|, \beta)$; the angle between them is $\theta = \beta - \alpha$
- (b)(c) In Cartesian coordinates, $\mathbf{a} = (\|\mathbf{a}\| \cos \alpha, \|\mathbf{a}\| \sin \alpha)$ and $\mathbf{b} = (\|\mathbf{b}\| \cos \beta, \|\mathbf{b}\| \sin \beta)$



Recovering the Algebraic Formula From Polar Coordinates

$$\cos(\beta - \alpha) = \cos \beta \cos \alpha + \sin \beta \sin \alpha \Rightarrow \mathbf{a} \cdot \mathbf{b} = a_1 b_1 + a_2 b_2$$

- In Cartesian coordinates, $\mathbf{a} = (\|\mathbf{a}\| \cos \alpha, \|\mathbf{a}\| \sin \alpha)$ and $\mathbf{b} = (\|\mathbf{b}\| \cos \beta, \|\mathbf{b}\| \sin \beta)$
- Substituting the expansion of $\cos(\beta - \alpha)$ into $\mathbf{a} \cdot \mathbf{b} = \|\mathbf{a}\| \|\mathbf{b}\| \cos \theta$ simplifies exactly to $a_1 b_1 + a_2 b_2$
- This shows the algebraic and geometric definitions describe the very same inner product, arrived at from two directions

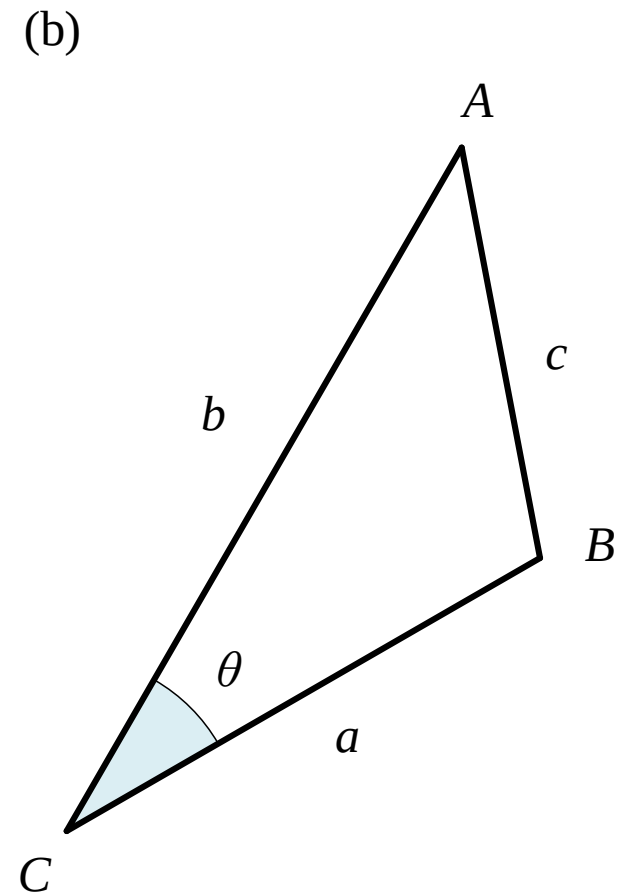
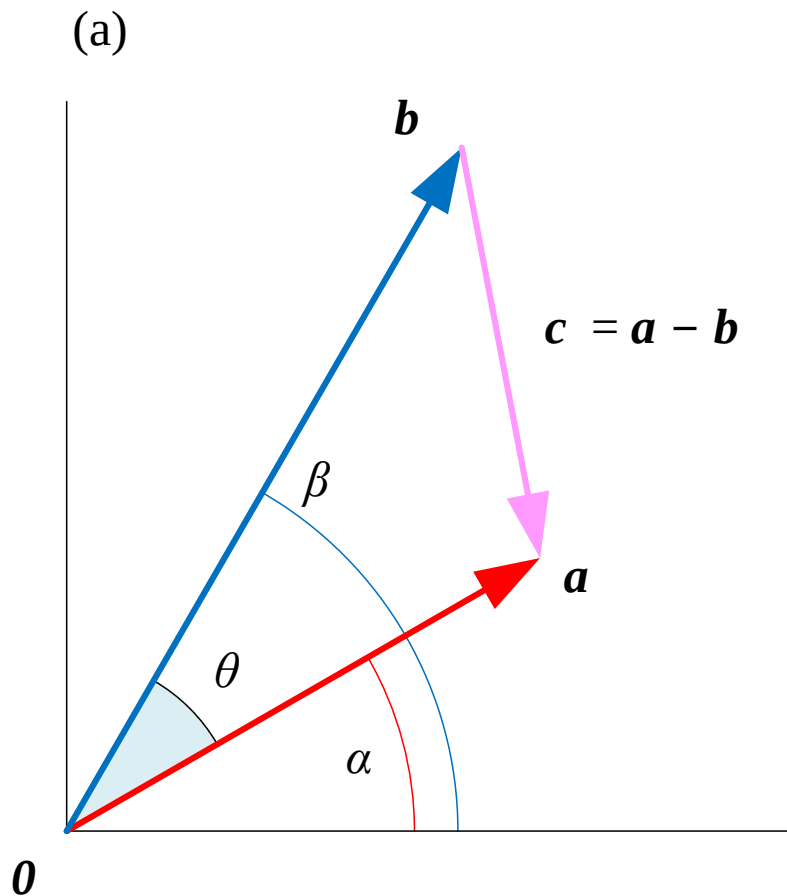
Deriving the Inner Product From the Law of Cosines

$$\|c\|^2 = \|a\|^2 + \|b\|^2 - 2\|a\|\|b\|\cos \theta, \text{ where } c = a - b$$

- Triangle side lengths a, b, c correspond to the lengths of $\mathbf{a}, \mathbf{b}, \mathbf{c}$, with θ the angle between sides a and b
- Introducing the vector difference $\mathbf{c}=\mathbf{a}-\mathbf{b}$, it can be shown that $\|c\|^2=\|a\|^2+\|b\|^2-2(\mathbf{a}\cdot\mathbf{b})$
- Comparing this with the law of cosines immediately gives $\mathbf{a}\cdot\mathbf{b}=\|a\|\|b\|\cos \theta$ — matching the geometric definition

Figure 10 · The Law of Cosines and the Inner Product

- (a) Introducing the vector difference $\mathbf{c} = \mathbf{a} - \mathbf{b}$, the three vectors $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ form a triangle
- (b) Triangle side lengths a , b , c correspond to the three vectors' lengths, with θ the angle between sides a and b



Key Terms in This Section

Inner Product $\mathbf{a} \cdot \mathbf{b} = \sum a_i b_i$ — a scalar result

Orthogonal Inner product equals 0 — geometrically, perpendicular

Orthonormal Vectors Pairwise orthogonal vectors, each with length 1

Acute / Obtuse Angle The angle ranges where the inner product is positive or negative

Polar Coordinates Describing a point or vector with (r, ϑ)

Scalar / Dot Product Two other common names for the inner product

Orthogonal Vectors A set of vectors with pairwise zero inner products

Scalar Projection $\|\mathbf{a}\| \cos \vartheta$ — the projected length of a vector along a direction

Cosine Similarity Using the cosine of the angle to measure how similar two vectors are

Law of Cosines The relation between a triangle's side lengths and the cosine of an angle

GO FURTHER

Companion Code and Further Reading

Code 1 · LA_01_05_01.ipynb

Compute and verify the inner product

Code 2 · LA_01_05_02.ipynb

Compute the cosine and angle between two vectors

Further Reading · Cross Product

<https://mathworld.wolfram.com/CrossProduct.html>

Open-Source Repository

All code and figures for the book: <https://github.com/Visualize-ML>

Q1 – Q10

Q1 Compute the inner product and angle for each pair, and explain the geometric meaning: $[1,0]^T, [0,1]^T$; $[-1,0]^T, [0,-1]^T$; $[1,0]^T, [-1,0]^T$; $[3,4]^T, [-4,3]^T$.

Q3 Given $\mathbf{x}=[x_1, x_2, x_3]^T$ orthogonal to $[1,1,1]^T$, find the relation among x_1 , x_2 , x_3 , and describe its geometric meaning.

Q5 Generate two random 3-D integer vectors with components in $[0,9]$ using Python, and compute their dot product and angle.

Q7 Study cosine similarity — its definition and where it is used.

Q9 Study the Hadamard product (element-wise product) of vectors on your own.

Q2 Given $\mathbf{x}=[x_1, x_2]^T$ orthogonal to $[1,1]^T$, find the relation between x_1 and x_2 , and describe its geometric meaning.

Q4 Write Python code to compute the pairwise inner products, cosines, and angles for the vectors in Figure 6.

Q6 Study how to compute the vector cross product on your own — see <https://mathworld.wolfram.com/CrossProduct.html>.

Q8 Preview the concepts of scalar projection and vector projection ahead of the next sections.

Q10 Write a function `inner_product(a, b)` that computes the inner product by hand (without `np.dot()`), and check it against `np.dot()`.

Q11 – Q20

Q11 Use Python to verify that commutativity, distributivity, and scalar associativity hold for random vectors.

Q13 Generate 10 random pairs of 2-D vectors, compute their angles, and classify each as acute, right, or obtuse.

Q15 Write a function `angle_between(a, b)` that returns the angle in degrees, handling the zero-vector case gracefully.

Q17 Compute the angle between white $[1,1,1]^T$ and each of red, green, blue, yellow, magenta, and cyan, and sort them by angle.

Q19 Generate the yellow, magenta, and cyan vectors from Figure 7 in Python, and verify their pairwise angles are all equal.

Q12 Write a function `is_orthogonal(a, b)` that tests whether two vectors are orthogonal (inner product ≈ 0 , with a tolerance).

Q14 Plot two vectors and their angle θ with Matplotlib; rotate one vector and observe how the sign of the inner product changes (as in Figure 2).

Q16 Use NumPy to verify that the RGB basis vectors are pairwise orthogonal, and check whether each is a unit vector.

Q18 Write a function `cosine_similarity(a, b)`, and use it to compare three random color vectors pairwise.

Q20 Verify in code that any vector $\mathbf{a}$ makes a 0° angle with itself and a 180° angle with $-\mathbf{a}$.

Q21 – Q29

Q21 Given polar coordinates $(\|a\|, \alpha) = (3, 30^\circ)$ and $(\|b\|, \beta) = (5, 90^\circ)$, use Python to compute the inner product of the two vectors.

Q23 Reproduce Figure 9 with Matplotlib, plotting two vectors a and b in polar form along with their angle $\theta = \beta - \alpha$.

Q25 Write a function `law_of_cosines_angle(a_len, b_len, c_len)` that recovers the angle θ from three triangle side lengths.

Q27 Write a function `vector_similarity_report(a, b)` that reports the inner product, angle in degrees, and cosine similarity together.

Q29 Use Python to verify that as dimension grows, two random vectors' angle tends toward 90° ("high-dimensional orthogonality"), and plot the trend.

Q22 Write a function `inner_product_from_polar(r1, alpha, r2, beta)` that computes the inner product from polar parameters, and check it against the Cartesian result.

Q24 Use Python to verify the law of cosines $\|c\|^2 = \|a\|^2 + \|b\|^2 - 2a \cdot b$ (with $c = a - b$) for random vectors.

Q26 Generate 100 random pairs of 3-D vectors in Python and report what fraction are orthogonal (inner product ≈ 0).

Q28 Plot 8 evenly spaced vectors on the unit circle with Matplotlib, compute all pairwise inner products, and visualize them as a heatmap.

10 Prompts to Go Deeper (1/2)

1

Explain why the sign of the inner product (positive, zero, negative) directly reflects how "similar" or "opposite" two vectors' directions are.

2

Prove the Cauchy-Schwarz inequality $|a \cdot b| \leq \|a\| \|b\|$, and describe the geometric condition under which equality holds.

3

Explain the intuition behind "orthogonal" using an everyday example, such as two unrelated variables.

4

Why does a zero inner product mean two vectors "share no common component"? Illustrate with the RGB color space.

5

Explain how cosine similarity is used in recommendation systems and natural language processing, such as with word vectors.

10 Prompts to Go Deeper (2/2)

6

Design a small Python experiment showing that in high dimensions, random vectors' angles tend to cluster near 90° .

7

If two vectors' inner product is negative, what does that mean geometrically and in practical applications?

8

Explain why "scalar product" emphasizes the algebraic view of the inner product, while "dot product" emphasizes its notation.

9

Using the law of cosines, discuss the connection between the inner product and triangle side lengths — what does it suggest about measuring distance?

10

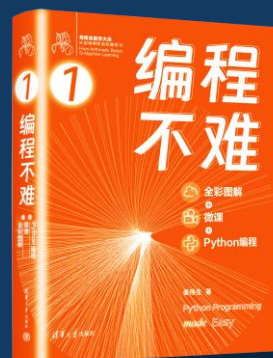
Describe a concrete application, such as image retrieval or matching user preferences, where the inner product or cosine similarity measures "how similar".

数学

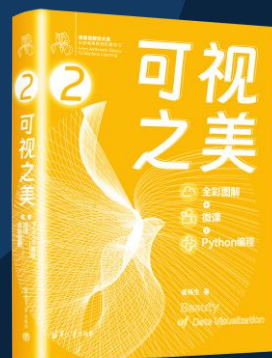
数学基础

线性代数

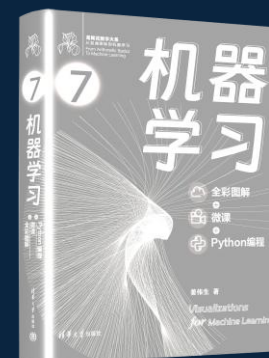
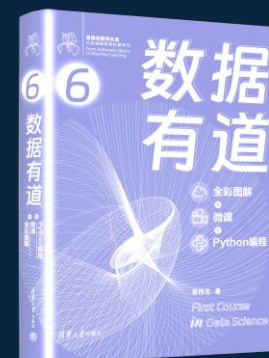
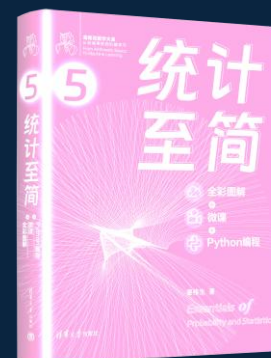
概率统计



Python编程



数据可视化



数据分析
图和网络

回归、降维
分类、聚类

工具

实践

